Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/31/2016
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Project Title:	Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	11/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 11/30/2017 per F. Hernandez/ARC (Ed., 10/24/16) NOTE: End date changed to 11/30/2016 per A. Chu/ARC (Ed., 12/10/15) NOTE: End date changed to 11/20/2015, per NSSC information (Ed., 8/10/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AM62G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE from June 2016 report: Nearly one year ago, we amended the grant proposal to include cryopreserved bovine and human semen, while eliminating the use of sea urchin and rat species. Sperm motility and acquisition of the capacity to fertilize are the most fundamental components of male reproduction necessary for species survival on Earth or in space. Previous space shuttle flight experiments that we performed demonstrated that the signal transduction (i.e.m protein phosphorylation) required for activation of sea urchin sperm occur more rapidly in microgravity than at 1G. In contrast, we found that signaling in sperm triggered by egg chemotactic factors is delayed in microgravity compared to 1G. No flight-capable microscope was available during those experiments to determine whether the signaling changes were in fact coupled to differences in motility under the microgravity versus 1G conditions. Our hypothesis is 1) that sperm motility (sea urchin and mammalian) is activated more rapidly in microgravity than on Earth, and 2) that the response of sea urchin and mammalian sperm (motility hyperactivation, and morphological changes) to fertilization activating conditions, as well as the underlying metabolic and regulatory mechanisms, are delayed or blocked in microgravity. In this regard, if the corresponding activation and changes in sperm motility do not occur, then fertilization cannot occur and reproduction will fail. Thus, it is essential to assess whether the critical component of sperm function, motility, and the underlying signalling and metabolic support are altered in space flight conditions. We propose to use existing flight hardware to activate sea urchin and rodent sperm in microgravity, and quantify multiple parameters of motility and morphology, combined with detailed metabolomic pathway analysis. Previously utilized methods for preparing inactive sea urchin sperm for subsequent activation in space flight will be adapted for mouse and rat sperm. We will also test whether activating peptides from sea urchin eggs, or mammalian capacitating conditions trigger temporal or magnitude changes in sperm motility, morphology, as well as essential regulatory glycolytic and energy metabolic pathway markers (using novel metabolomic analysis) in microgravity versus 1G. Most importantly, this comprehensive project will lead to the first direct in vitro assessment of canonical fertility-critical functional processes of mammalian sperm in microgravity that are identical to many human in vitro clinical and research laboratory assessments. These experiments are 1) part of the Rapid Turn-around Space Flight Experiments Research Emphasis, and 2) address the National Research Council (NRC) 2011 Decadal Survey to discover and characterize underlying mechanisms of adaptation to the space flight environment, and to determine cellular and organismal mechanisms that regulate and sustain reproduction and development in space. In this regard, the results of these experiments will provide important fundamental reproductive biological data to support NASA's goal to conduct multigenerational studies of mammals in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous studies indicate that reproductive functions in sperm that occur on the ground also occur in space flight. However, the timing of signal transduction changes upon initiation of sperm motility is more rapid, yet, the fertility-related responses of sperm to the presence of egg chemotactic factors are delayed. This suggests that sperm fertility may be affected resulting in subfertility or infertility in males during and post flight. There is insufficient data available from humans to assess this possibility due to low n's and HIPPA concerns. The flight logistics for the new SpaceX vehicle require a wider window of time between sample handover and subsequent experiment activation by crew on the International Space Station (ISS). New procedures for extending the viability of quiescent sperm are being developed for both sea urchins and rodents (rats).
Task Progress:	In the past year of research, we have made considerable progress. Nearly one year ago, we amended the grant proposal to include cryopreserved bovine and human semen, while eliminating the use of sea urchin and rat species. This was a significant alteration to the project, one that required both learning about cryopreservation and how to utilize samples preserved in this way. Particularly, we needed to examine the process of cryopreservation and the various media used in that process in order to determine any effects on existing experimental protocols for flight. We found that while bovine and human cryopreserved samples are more efficient with regard to flight constraints and are more translational, their use adds considerable technical challenges to the otherwise straightforward experiments. One such complication is the existence of cryopreservatives, which act antagonistically to the media used to induce capacitation in the sperm. We learned that the proprietary semen extender used to dilute the cryopreserved bovine samples is based on a Tris-Citrate-Fructose media, and citrate being a calcium chelator, sequesters the exogenous calcium added to our capacitation-inducing media, effectively reducing our ability to cause such an induction. Because of this it was necessary to determine the extent to which exogenous calcium was reduced due to this effect. Since chelation effects are dependent on many different conditions and cannot be modeled in a simple manner, we utilized a specific open-source computer program to calculate percent calcium chelation at variable total calcium concentrations in the capacitation medium (CM). In this way, we were able to construct a chelation curve for our CM, which demonstrated the chelating effect of citrate. We determined that we needed to supplement the CM with 8 mM calcium chloride, which then is reduced to 2 mM, the desired calcium chloride concentration for the CM. We are currently in the process of verifying this approach experimentally by adding variable concentrations of calcium to the CM and assessing motility for levels of capacitation-associated motility using computer aided semen analysis (CASA). Additionally, we have successfully demonstrated the general efficacy of our CM at inducing capacitation as compared to a non-capacitation-inducing control media (NCM). These media formulations were adapted from extensive literature research and are effective in each of their respective species: rat, bovine, and human as demonstrated through motility data. We also conducted long duration sperm incubations to determine the minimum amount of time required before the capacitation state is evident in the sperm. With regard to the flight motility analysis, we have arranged flight hardware adaptability for these experiments, specifically the microscope and analysis programming used. We have coordinated with the software programmers at MofA Global to enable us to export video data from the ISS to our ground computers for analysis, a vital logistical step in flight definition. Although in the past year we experienced catastrophic loss of our primary CASA machine, we have substituted our secondary CASA system in its place. We are currently in the process of setting up our current system to be able to accurately analyze video imported from the ISS. Further coordinating with BioServe Space Technologies, we have been determining experimental requirements for space flight hardware in which we will be able to incubate the sperm and perform microscopic video analysis. CASA analysis requires around a 50 cell per field of view sperm concentration to accurately measure motility parameters, and so it was necessary to determine the final sperm concentration (M/ml) that yielded this cell density. We devised experiments testing variable concentrations of sperm of each species to determine the concentration at which we achieved this 50 cell requisite. Using these optimal final sperm concentrations, in addition to the principal sperm concentrations of the cryopreserved samples, we were able to make calculations for determining the necessary volume of the incubation chambers used for the video capture, while

	minimizing the amount of cryopreservant present in the overall incubation. These hardware constraints were communicated to BioServe and will be vital in their prototyping of the space flight hardware. The other aspect of the project is the examination of molecular biomarkers that reflect the acquisition of fertilizing capacity. It has long been known that tyrosine phosphorylation is instrumental in the process of capacitation, although very few proteins have been specifically identified. We began by examining global protein tyrosine phosphorylation of the entire sperm cell lysate obtained under conditions that would be amenable to the logistical constraints of space flight. We found that because we supplement the CM and NCM with a high concentration (6000 µg/ml) of bovine serum albumin (BSA), the western immunoblotting analysis showed a very prominent BSA band at 65 kDa in addition to numerous others resulting from protein degradation, which obscured many of the band sizes described in the literature. Also, measuring a single band on a western immunoblot of global tyrosine phosphorylation was nearly impossible to accomplish as optical density analysis software takes into account background surrounding the band in question, an area which contained proteins that may or may not be phosphorylated in the same manner, causing an unacceptable degree of statistical noise. To remedy this we resolved to identify known proteins that have been demonstrated to become tyrosine phosphorylated in sperm upon capacitation. This however is a tricky proposition given that it would need to be a protein that is conserved across rat, bovine, and human species, as well as a protein that is phosphorylated in the same way since it is known that the same protein may become phosphorylated differently between species. We also required a protein that was expressed in high enough levels in sperm. From extensive literature research, we have found a small number of proteins that exhibit these qualities: A-kinase anchoring protein 82 (AKAP82), heat shock protein 90 (HSP-90), tektin-2, and calcium-binding tyrosine phosphorylation-regulated protein (CABYR). The question then became how to examine the phosphorylation state of these proteins in isolation such that their optical density data retained a high degree of fidelity. The solution to this was to use immunoprecipitation (IP) to precipitate out each of these proteins from the whole sperm cell lysate and then utilize 4G10 phosphotyrosine antibody to examine the phosphorylation state of each respective protein. Once a reliable IP protocol is established for these cells, we will look at how the behavior of these known molecular biomarkers relates to the motility data. Together these two methods will provide a strong indication of the state of capacitation in sperm. To develop such a protocol, we needed to know the total protein content of the samples. Generally, spectrophotometry is used to determine this, but as this is an inviable approach in this instance due to the high BSA concentration in the incubation media, we needed an alternate method. Instead we harvested the caudal epididymides from a rat and fixed in 4% paraformaldehyde. We then made 50% serial dilutions down to 1/64 of the stock. Spectrophotometric analysis was used to then determine the protein content of each of these dilutions. Then, the sperm cell concentration of each dilution was determined via hemocytometer count. Together these data created a sperm concentration (M/ml) v. protein content (ug/ml) curve, which will allow us to more refine the IP and western immunoblotting protocols. We have also been working closely with the research directors of both SelectSires, Inc. and California Cryobank to arrange sourcing for bovine and human semen samples, respectively, that meet experimental quality and logistical requirements for flight.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Tash JS, Okwuazi EM, Yarns EM. "Suitability of Human and Bovine Sperm for In Vitro Reproductive Biology Studies on the ISS." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. https://asgsr.org/index.php/2015-abstracts , Nov-2015